

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, Java, SQL, JavaScript, C++, Bash

Tools/Frameworks: PyTorch, TensorFlow, Hugging Face, LangChain, FastAPI, Flask, Kubernetes, Docker, TensorRT, Azure, AWS, GCP, Azure Databricks

Competence: Machine Learning, Deep Learning, Generative AI, MLOps, DevOps, Cloud-Native Development, NLP, Semantic Knowledge Representation, Model Optimization, Multi-Agent Systems

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated deep learning models for metrological applications, achieving improved model performance and training efficiency.
- Implemented novel architectures and physics-informed hybrid loss functions, enhancing model accuracy and stability.
- Collaborated with PhD researchers to publish findings at the FLOMEKO conference and developed dashboards for data visualization and model performance monitoring.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Enhanced vitiligo segmentation accuracy to 94% and improved inference speed by 5x using advanced AI algorithms, deployed via Docker and FastAPI.
- Developed a real-time tennis foul detection system leveraging TensorRT, reducing compute cost and inference time by 6x.
- Managed MLOps pipelines on AWS SageMaker, resolved YOLO model data drift, and maintained Kubernetes infrastructure for scalable AI deployment.
- Designed and delivered educational sessions on machine learning and big data to university audiences, fostering knowledge sharing.
- Built POCs using YOLO, few-shot learning, and CNNs to demonstrate AI solutions for client needs.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Developed a full-stack web application for AI-based services using Flask, HTML, and CSS.
- Built an AI chatbot leveraging OpenAI API and state-of-the-art LLM tools to enhance user engagement and automation.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Designed and implemented web scrapers and ETL pipelines for structured and unstructured data processing.
- Automated data ingestion and cleaning workflows, enabling efficient client analytics and reporting.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Developed a weighted-probability ensemble integrating DPFunc and DeepGOPlus, leveraging AlphaFold for feature extraction and taxonomic bias mitigation.
- Achieved a 4% accuracy improvement through custom feature engineering across molecular function, cellular component, and biological process namespaces.

Razanlytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Implemented a real-time Bitcoin price forecasting system using advanced time-series analysis and machine learning techniques.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Developed the world's first AI-driven self-playing Chitralli Sitar by integrating signal processing and music generation models.
- Project featured on Coke Studio in a music video by Hassan Raheem, showcasing innovative AI applications in music.

Hugging Face

Open-Source

Contributor

- Co-authored the Zero-shot Image Classification technical documentation and implementation page on the Hugging Face Hub.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Specialized in Machine Learning, Deep Learning, and Generative AI with a focus on cloud-native development and MLOps.

National University of Computer and Emerging Sciences Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Graduated with honors and recognized on the Dean's List of Honor for academic excellence.
- Completed coursework in Data Structures, Algorithms, Machine Learning, and Artificial Intelligence.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python